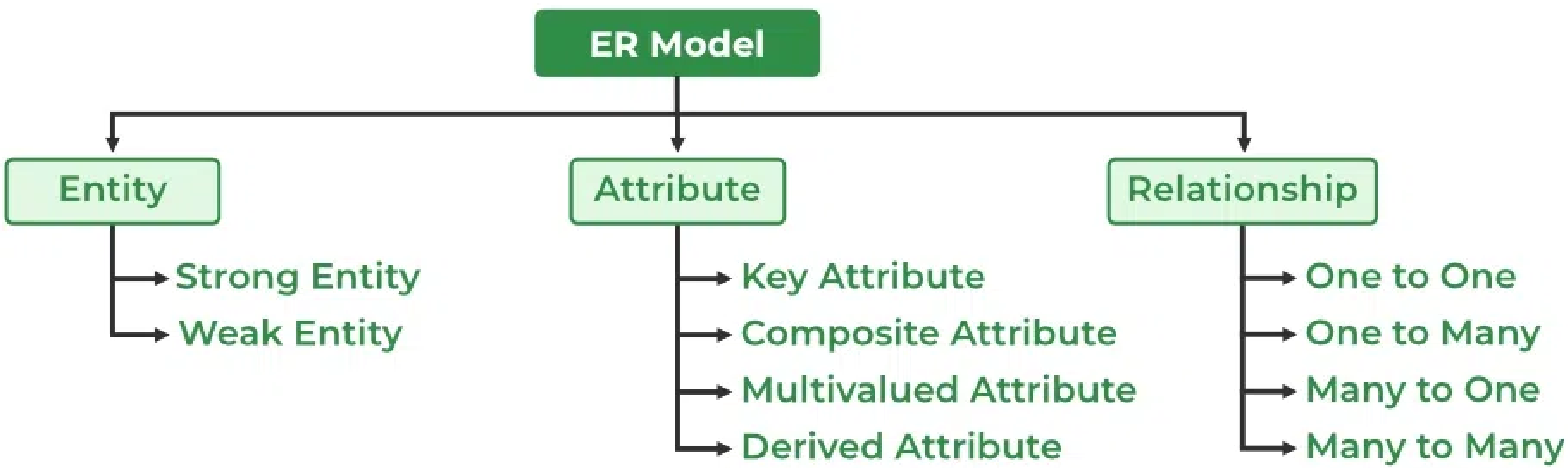
Lab 7

Introduction to Databases

Entity-Relationship Model (ER Model)

The Entity-Relationship Model (ER Model) is a conceptual model for designing a database. This model represents the logical structure of a database, including entities, their attributes, and relationships between them.

Entity: An object that is stored as data. E.g: *Student*, *Course,* or *Company*.
Attribute: Properties that describe an entity. E.g: *StudentID, CourseName, or EmployeeEmail.*
Relationship: A connection between entities. E.g: *Student* enrolls in a *Course*.



The graphical representation of this model is called an Entity-Relation Diagram (ERD).

ER Model in Database Design Process

We typically follow the below steps for designing a database for an application.

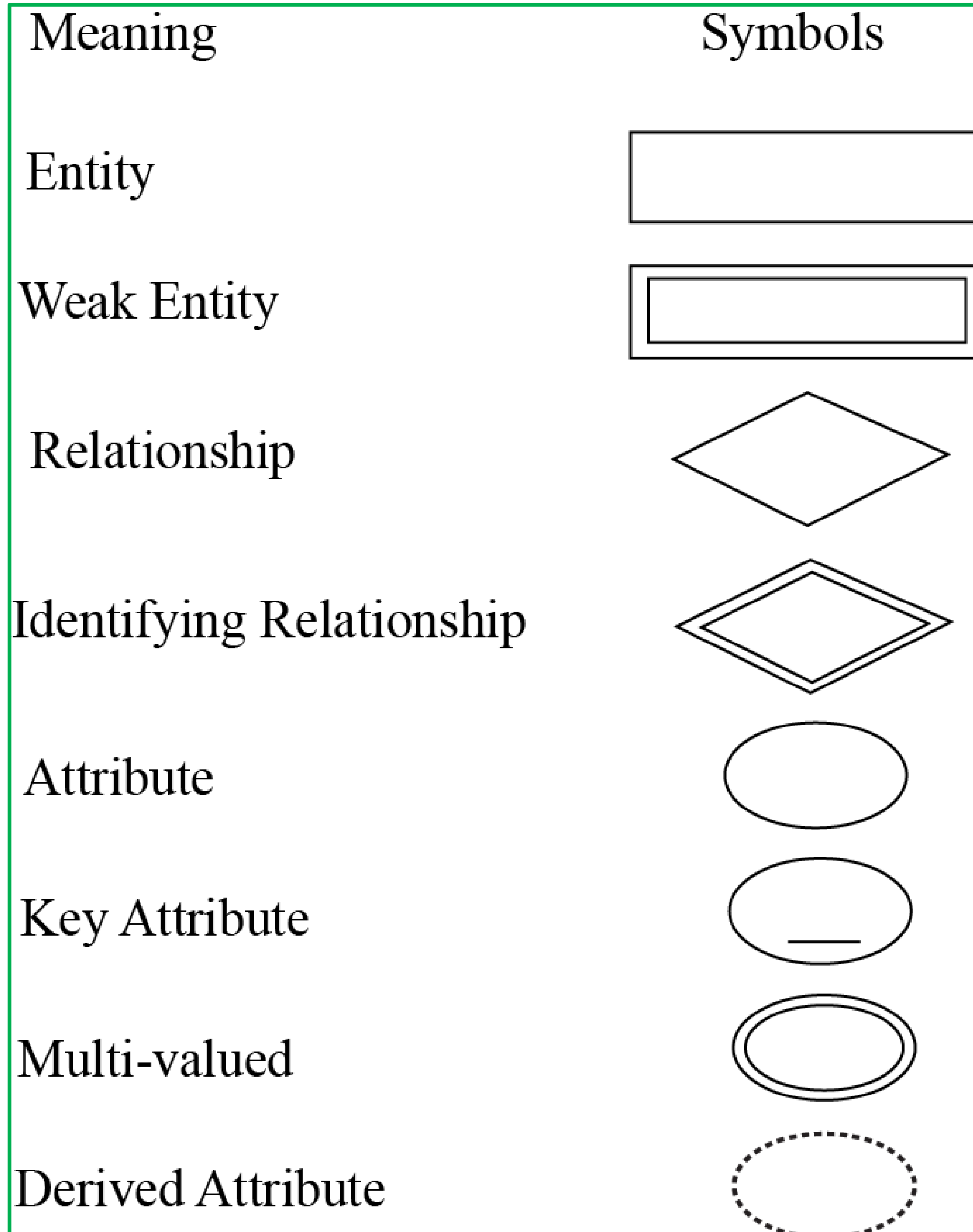
1. Gather the requirements (functional and data) by asking questions to the database users.
2. Create a logical or conceptual design of the database. This is where ER model platheys a role. It is the most used graphical representation of the conceptual design of a database.
3. After this, focus on physical database design (like indexing) and external design (like views).

Uses of ER Diagrams in DBMS

- ER diagrams represent the E-R model in a database, making them easy to convert into relations (tables).
- These diagrams serve the purpose of real-world modeling of objects which makes them intently useful.
- Unlike technical schemas, ER diagrams require no technical knowledge of the underlying DBMS used.
- They visually model data and its relationships, making complex systems easier to understand.

Symbols Used in ER Model

ER Model is used to model the logical view of the system from a data perspective which consists of these symbols:



Entity

It represents a real-world object, concept or thing about which data is stored in a database. It act as a building block of a database. Tables in relational database represent these entities.

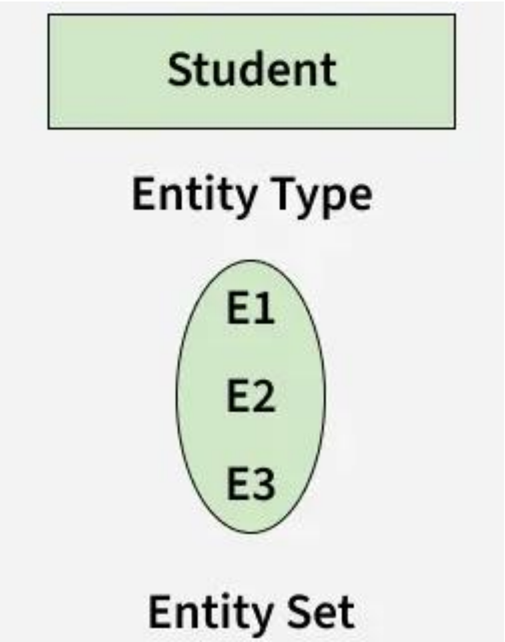
Example of entities:

- **Real-World Objects:** Person, Car, Employee etc.
- **Concepts:** Course, Event, Reservation etc.
- **Things:** Product, Document, Device etc.

The entity type defines the structure of an entity, while individual instances of that type represent specific entities.

Entity Set

Refers to an individual object of an entity type, and the collection of all entities of a particular type is called an entity set. For example, E1 is an entity that belongs to the entity type "Student," and the group of all students forms the entity set.



Types of Entity

There are two main types of entities:

1. **Strong Entity**
   A type of entity that has a key Attribute that can uniquely identify each instance of the entity. A Strong Entity does not depend on any other Entity in the Schema for its identification. It has a primary key that ensures its uniqueness and is represented by a rectangle in an ER diagram.

2. **Weak Entity**
   It cannot be uniquely identified by its own attributes alone. It depends on a strong entity to be identified. A weak entity is associated with an identifying entity (strong entity), which helps in its identification. A weak entity are represented by a double rectangle. The participation of weak entity types is always total. The relationship between the weak entity type and its identifying strong entity type is called identifying relationship and it is represented by a double diamond.

Attributes in ER Model

[Attributes](#)are the properties that define the entity type. For example, for a Student entity Roll_No, Name, DOB, Age, Address, and Mobile_No are the attributes that define entity type Student. In ER diagram, the attribute is represented by an oval.

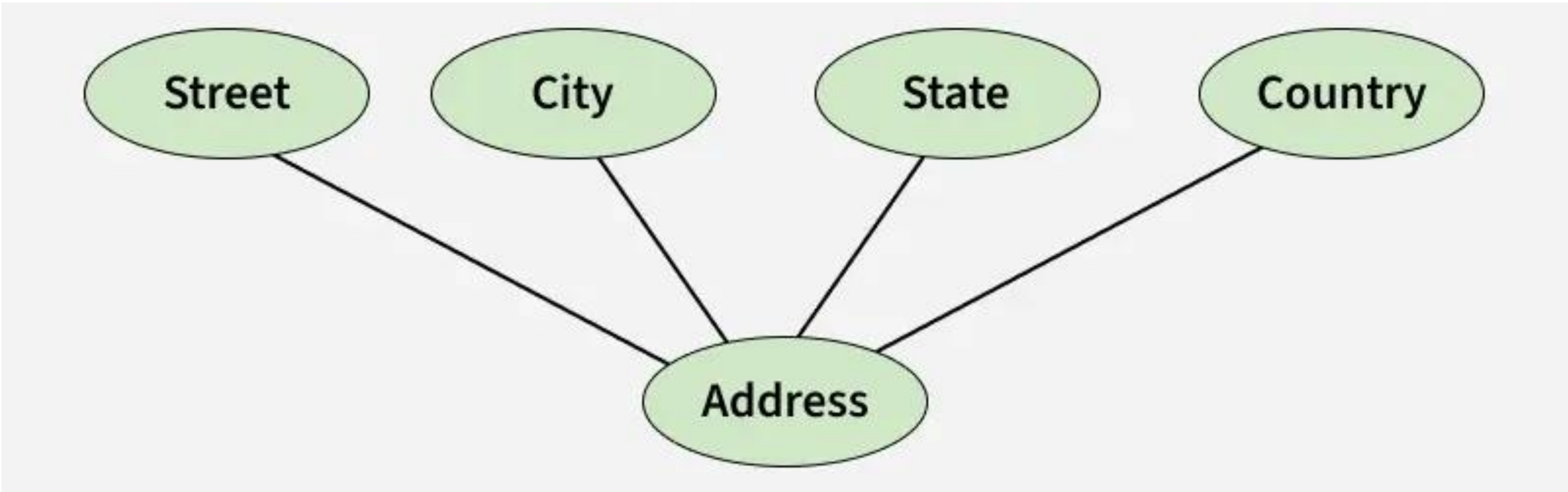
Types of Attributes

1. Key Attribute

The attribute which uniquely identifies each entity in the entity set is called the key attribute. For example, Roll_No will be unique for each student. In ER diagram, the key attribute is represented by an oval with an underline.

2. Composite Attribute

An attribute composed of many other attributes is called a composite attribute. For example, the Address attribute of the student Entity type consists of Street, City, State, and Country. In ER diagram, the composite attribute is represented by an oval comprising of ovals.

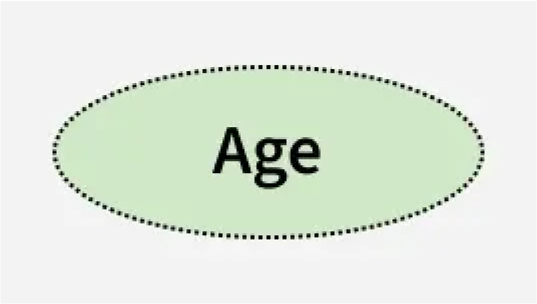


3. Multivalued Attribute

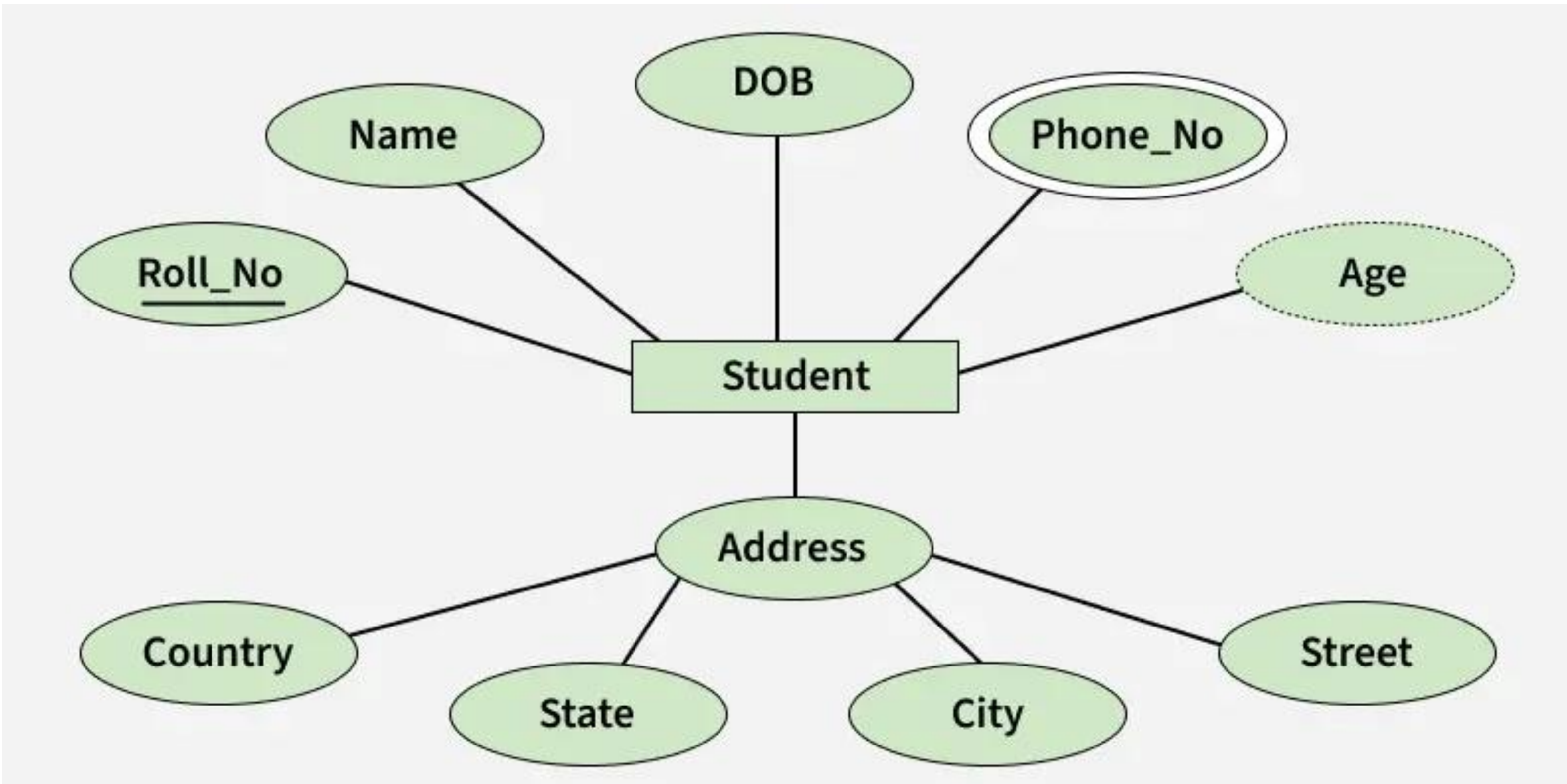
An attribute consisting of more than one value for a given entity. For example, Phone_No (can be more than one for a given student). In ER diagram, a multivalued attribute is represented by a double oval.

4. Derived Attribute

An attribute that can be derived from other attributes of the entity type is known as a derived attribute. e.g.; Age (can be derived from DOB). In ER diagram, the derived attribute is represented by a dashed oval.

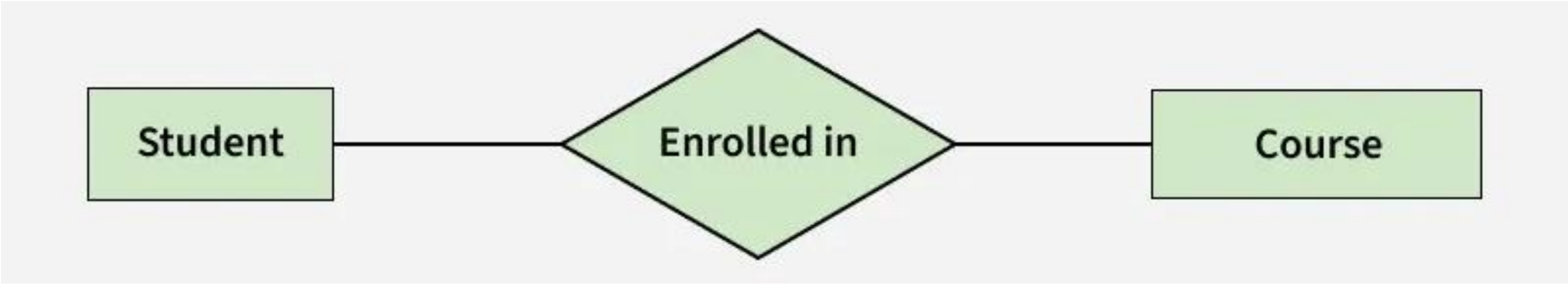


The Complete Entity Type Student with its Attributes can be represented as:

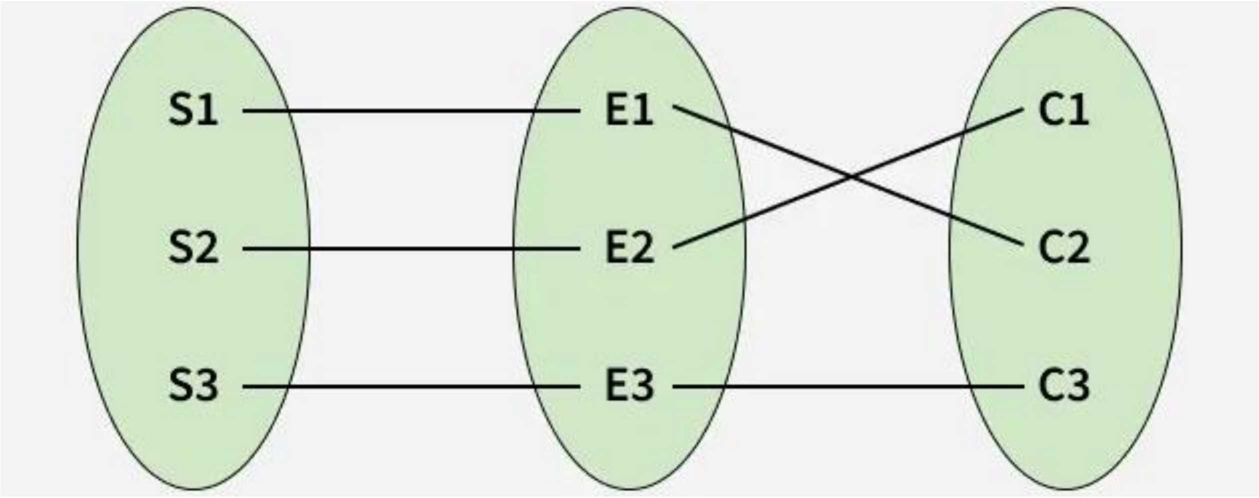


Relationship Type and Relationship Set

A Relationship Type represents the association between entity types. For example, ‘Enrolled in’ is a relationship type that exists between entity type Student and Course. In ER diagram, the relationship type is represented by a diamond and connecting the entities with lines.



A set of relationships of the same type is known as a relationship set. The following relationship set depicts S1 as enrolled in C2, S2 as enrolled in C1, and S3 as registered in C3.



Degree of a Relationship Set

The number of different entity sets participating in a relationship set is called the degree of a relationship set.

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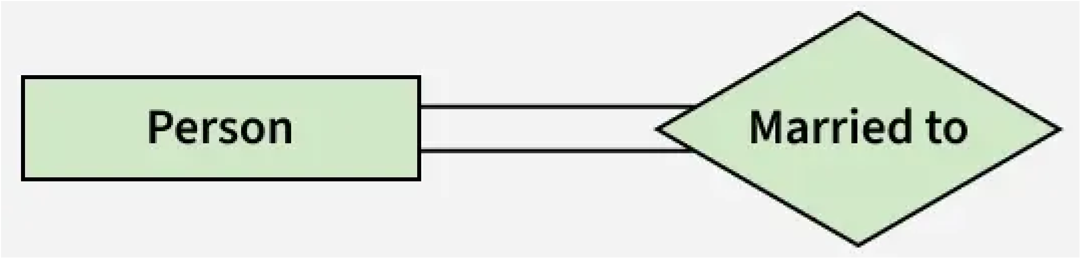
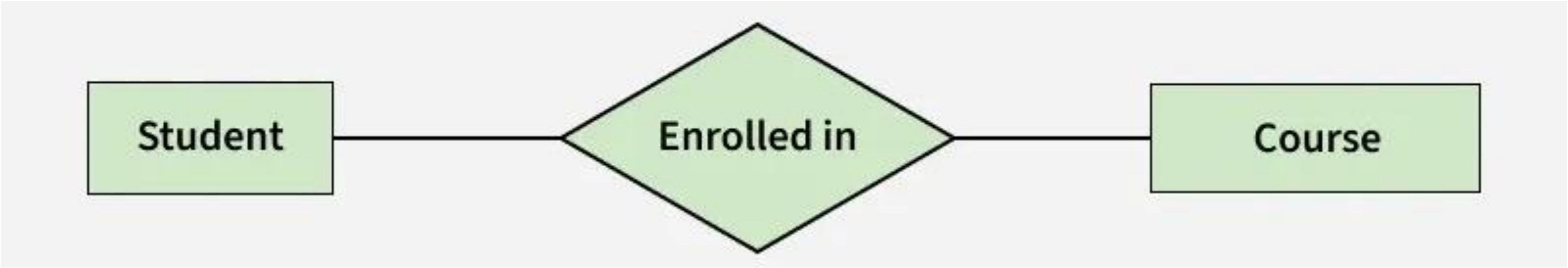
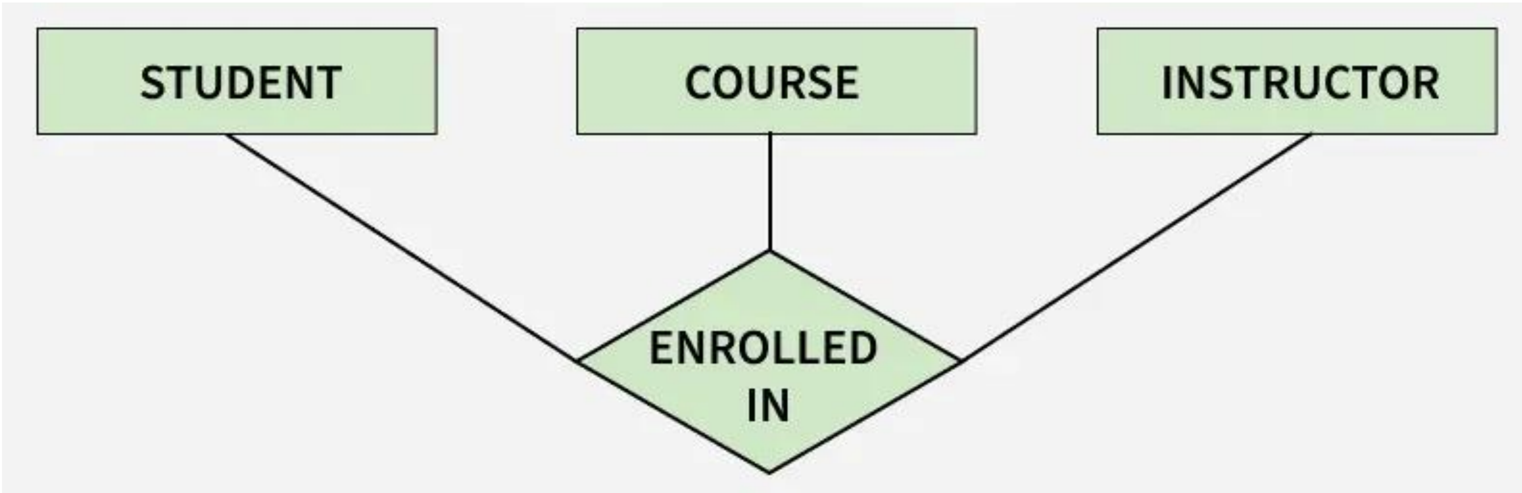
1. **Unary/Recursive Relationship:** When there is only ONE entity set participating in a relation. For example, one person is married to only one person



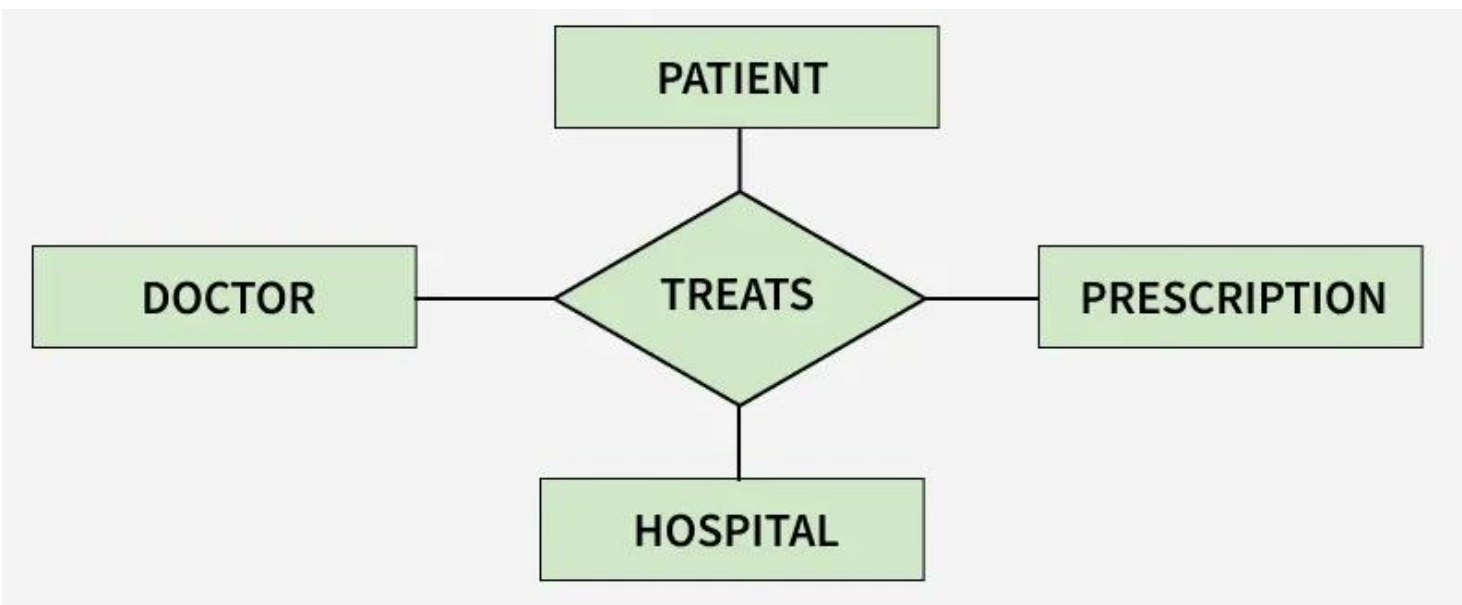
2. **Binary Relationship:** When there are TWO entities set participating in a relationship. For example, a Student is enrolled in a Course.



3. Ternary Relationship: When there are three entity sets participating in a relationship.



4. N-ary Relationship: When there are n entities set participating in a relationship, the relationship is called an n-ary relationship.

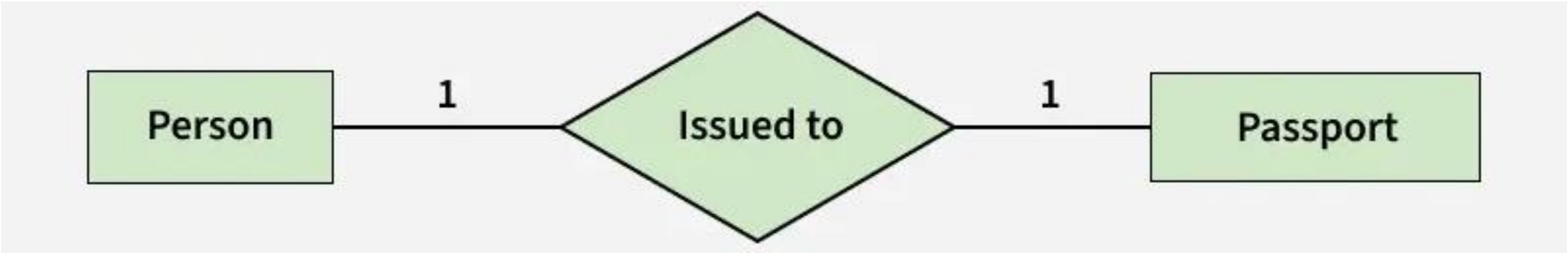


Cardinality in ER Model

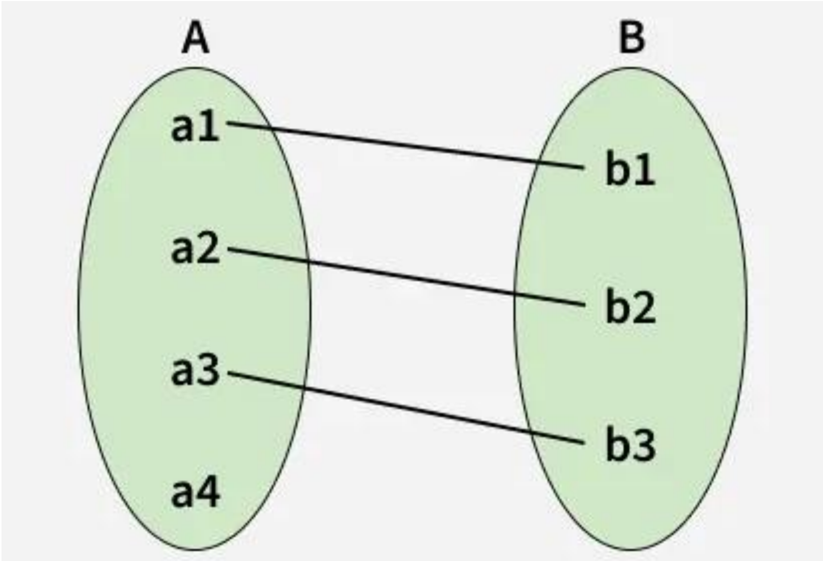
The maximum number of times an entity of an entity set participates in a relationship set is known as cardinality. Cardinality can be of different types:

1. One-to-One

When each entity in each entity set can take part only once in the relationship, the cardinality is one-to-one. Let us assume that one person can be issued only one passport, and one passport is issued to only one person. So, the relationship will be One-to-One (1 : 1), meaning that each person has a single passport, and each passport belongs to a single person.

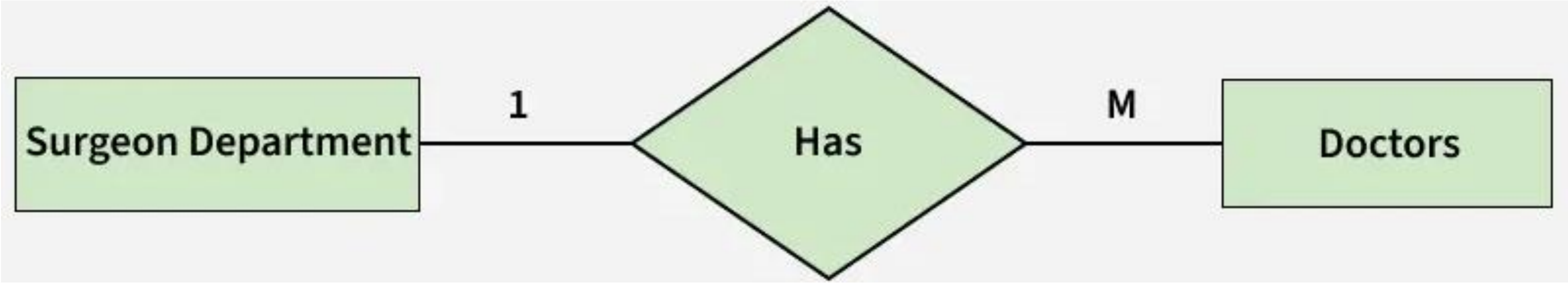


Using Sets, it can be represented as:

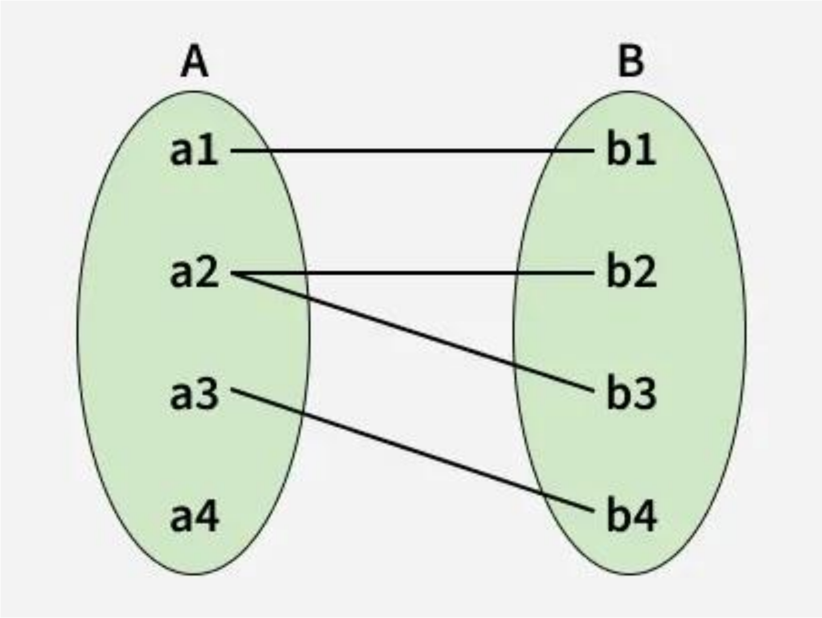


2. One-to-Many

In a one-to-many relationship, one entity can be associated with multiple entities. For example, a single Surgeon Department can have many Doctors. Therefore, the cardinality of this relationship is 1 to M, meaning one department can have many doctors.



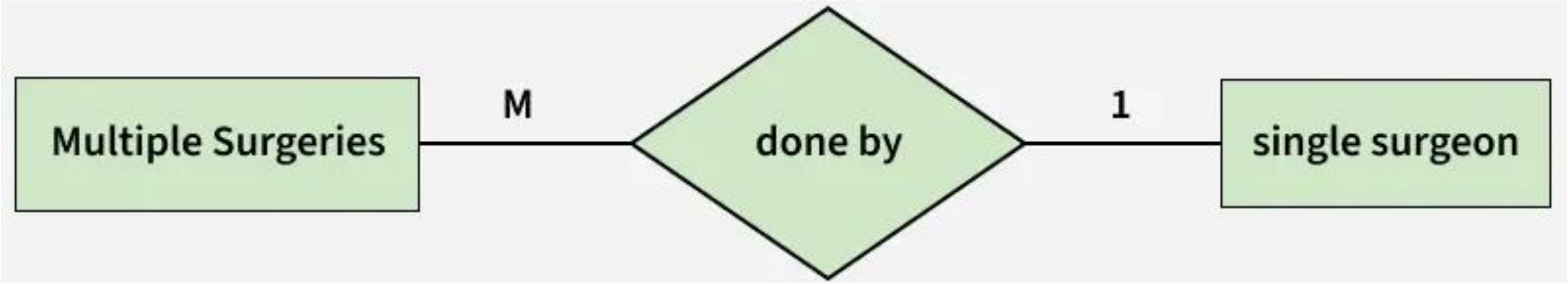
Using sets, one-to-many cardinality can be represented as:



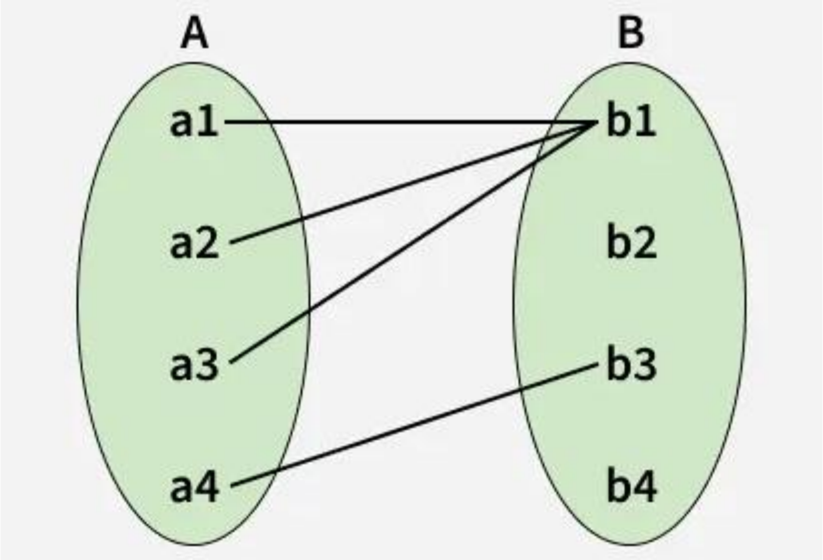
3. Many-to-One

When entities in one entity set can take part only once in the relationship set and entities in other entity sets can take part more than once in the relationship set, cardinality is many to one.

Let us assume that multiple surgeries can be performed by one surgeon, but one surgery is performed by only one surgeon. So, the cardinality will be M to 1, meaning that many surgeries can be done by a single surgeon, but each surgery is done by only one surgeon.

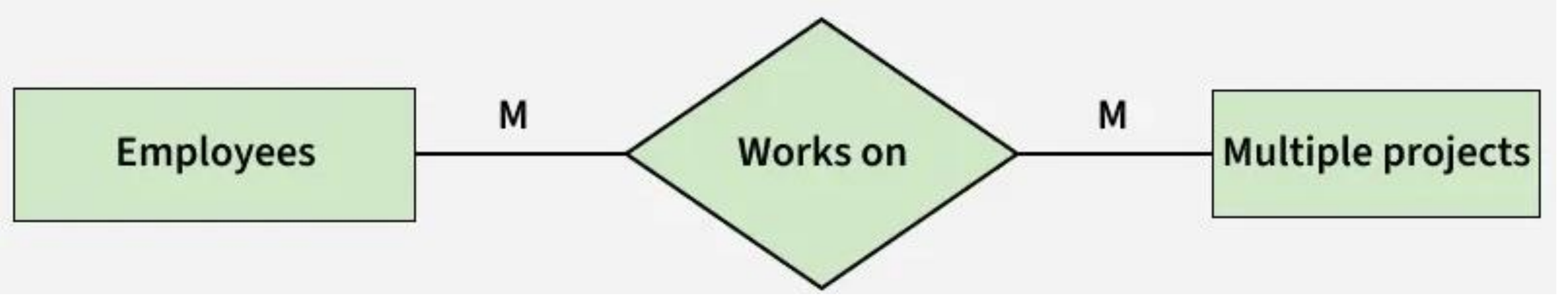


Using Sets, it can be represented as:

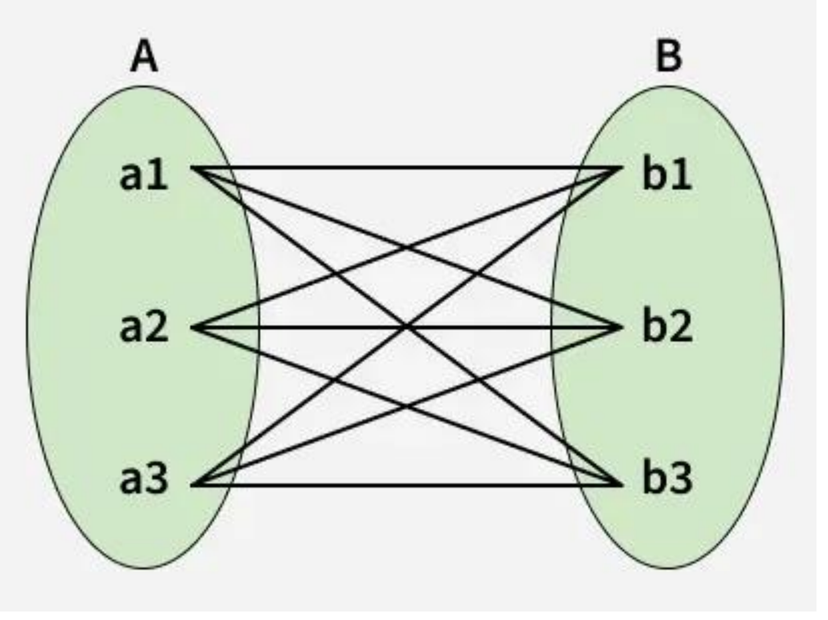


4. Many-to-Many

When entities in all entity sets can take part more than once in the relationship cardinality is many to many. Let us assume that an employee can work on multiple projects and each project can have multiple employees working on it. So, the relationship will be many-to-many (M:N), meaning that one employee may be associated with several projects, and one project may involve several employees.



Using Sets, it can be represented as:



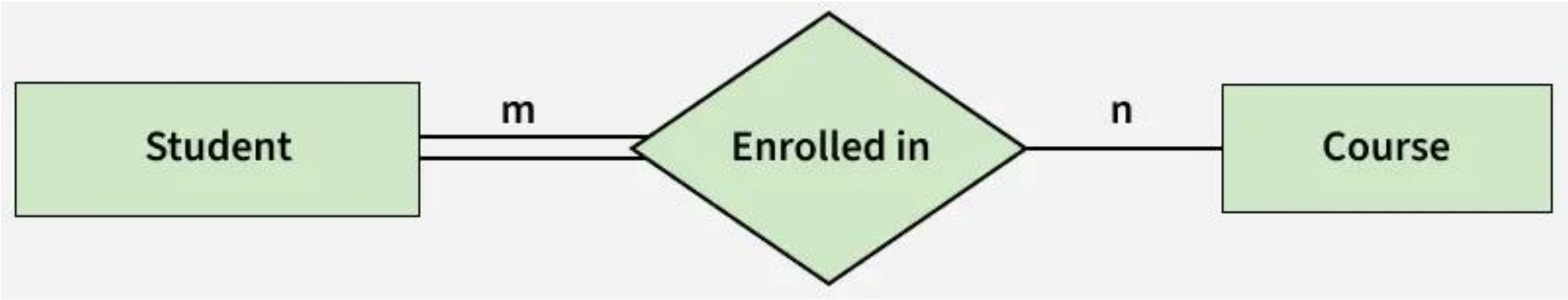
Participation Constraint

Participation Constraint is applied to the entity participating in the relationship set.

1. **Total Participation:** Each entity in the entity set must participate in the relationship. If each student must enroll in a course, the participation of students will be total. Total participation is shown by a double line in the ER diagram.

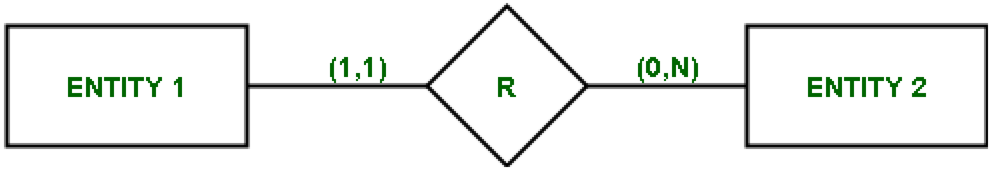
2. **Partial Participation:** The entity in the entity set may or may NOT participate in the relationship. If some courses are not enrolled by any of the students, the participation in the course will be partial.

The diagram depicts the 'Enrolled in' relationship set with Student Entity set having total participation and Course Entity set having partial participation.



Structural Constraints

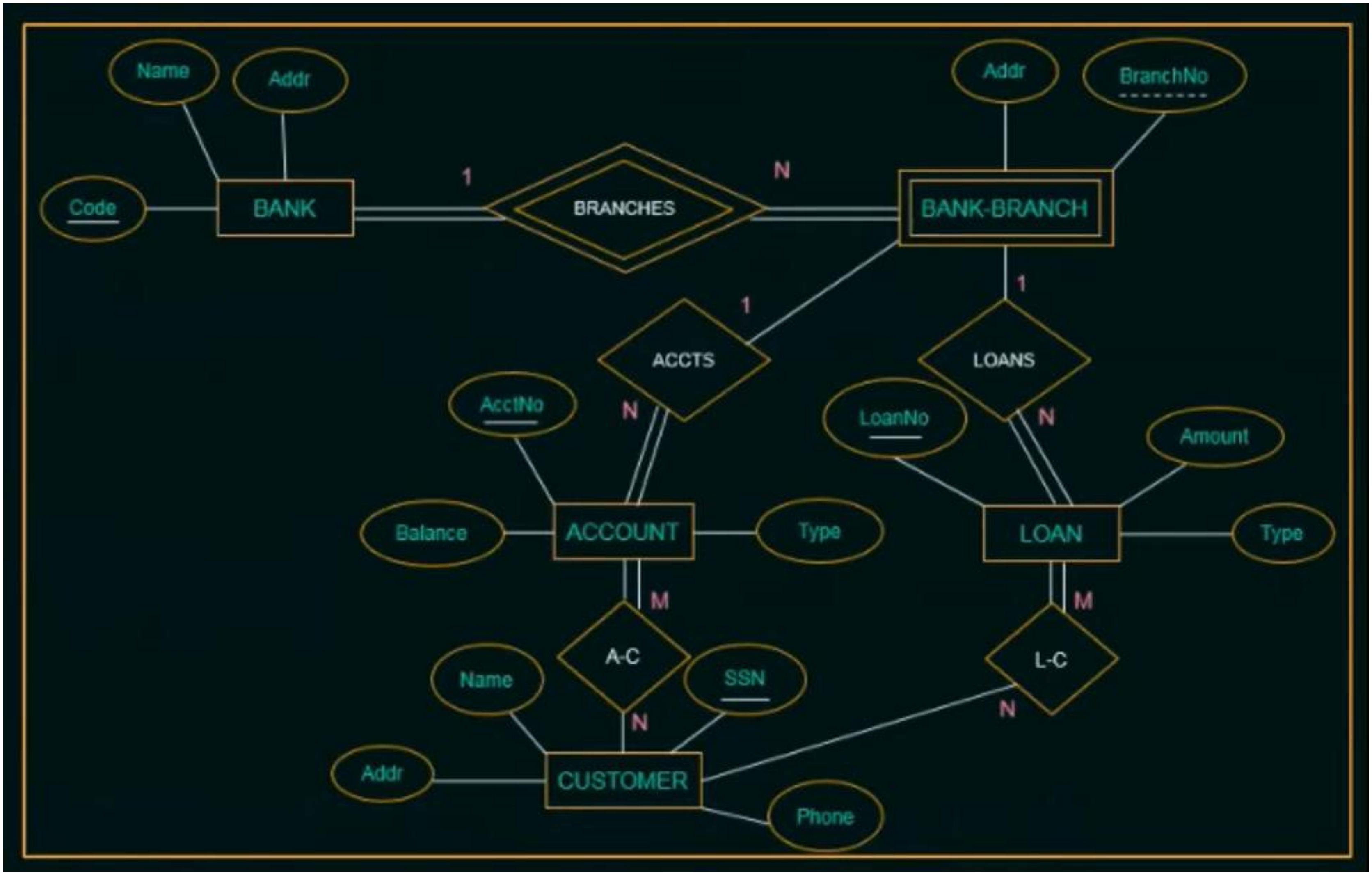
Structural Constraints are also called Structural properties of a [database management system (DBMS)](). Cardinality Ratios and Participation Constraints taken together are called Structural Constraints. The name constraints refer to the fact that such limitations must be imposed on the data, for the DBMS system to be consistent with the requirements.



The Structural constraints are represented by **Min-Max notation**. This is a pair of numbers(m, n) that appear on the connecting line between the entities and their relationships. The minimum number of times an entity can appear in a relation is represented by m whereas, the maximum time it is available is denoted by n. If m is 0 it signifies that the entity is participating in the relation partially, whereas, if m is either greater than or equal to 1, it denotes total participation of the entity

Exercises

Problem 1:



Consider the ER diagram shown in the figure for part of a BANK database. Each bank can have multiple branches, and each branch can have multiple accounts and loans.

a) List the nonweak entity types in the ER diagram.

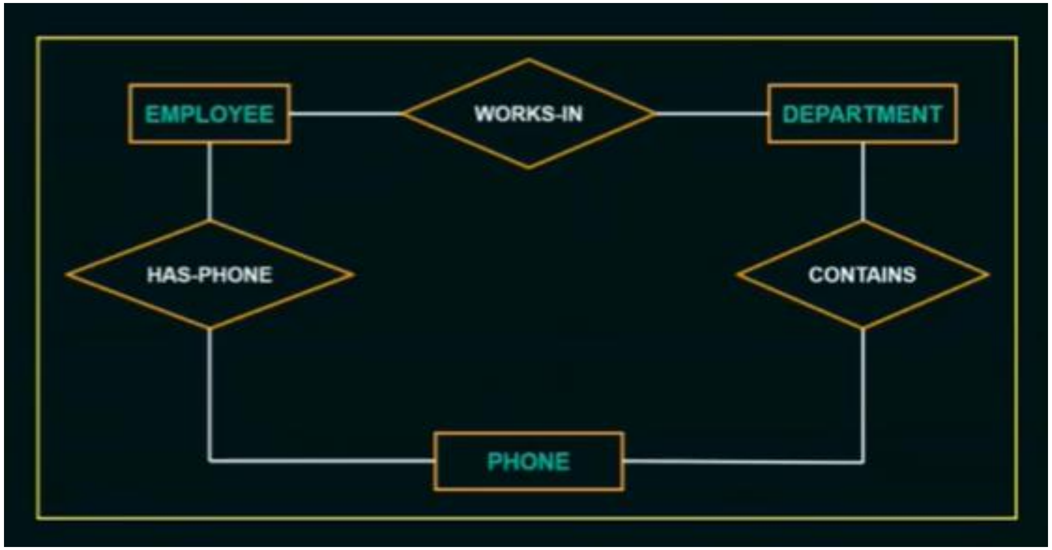
b) Is there a weak entity type? If so, give its name, partial key, and identifying relationship

c) List the names of all relationship types, and specify the (min,max) constraint on each participation of an entity type in a relationship type, justify your choices

d) Suppose that every customer must have at least one account but is restricted to at most two loans at a time, and that a bank branch cannot have more than 1000 loans. How does this show up on the (min, max) constraints?

Problem 2:

Consider the ER diagram in the figure. Supply (min,max) constraints on this diagram.



Assume that :

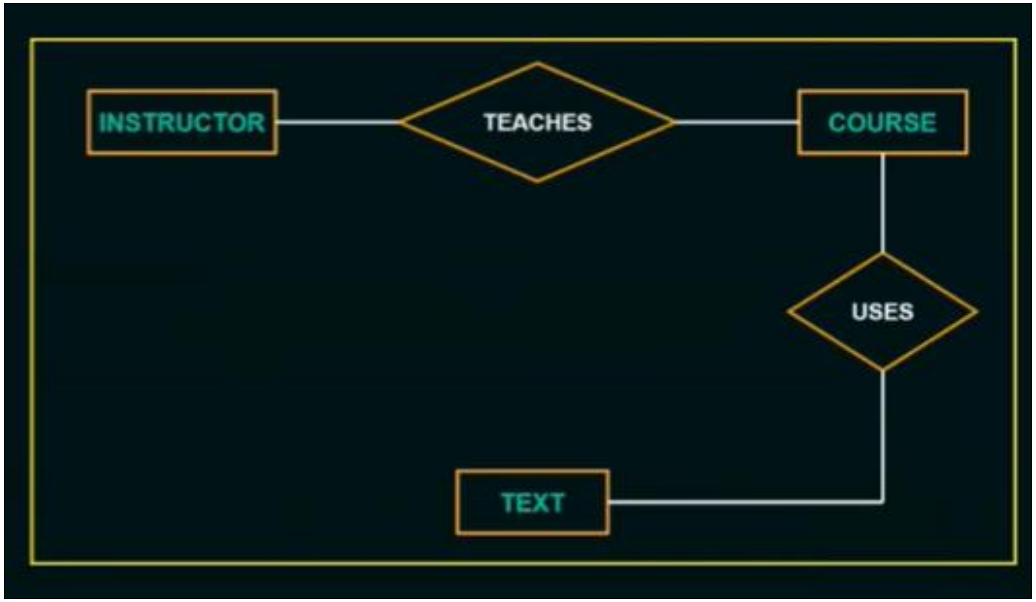
- An employee may work in upto two departments or may not be assigned to any department.
- Each department must have one and may have up to three phone numbers
- Each department can have between 1 to 10 employees.
- Each phone is used by only one department.
- Each phone is assigned to at least one employee and maybe assigned to upto 5 employees.
- Each employee must have at least one phone and may have upto 3 phones.

Problem 3:

Consider the ER diagram in the figure. Assume that :

- A course may or may not use a textbook, but that a text by definition is a book tht is used in some course.
- A course may not use more than five books.
- Instructors teach from two to four courses.
- Each course is taught by exactly one instructor.
- Each textbook is used by only one course.
- An instructor dose not have to adopt a textbook for all courses.
- If a text exists, (a) it is used in some course (b) it is adopted by a instructor who teaches that course.

Supply (min,max) constraints on this diagram. If we add the relationship ADOPTS between INSTRUCTOR and TEXT, what (min,max) constraints would you put on it?why?



Problem 4:

Draw a complete ER diagram for University Management System , showing all entities, attributes, relationships, and cardinalities. Indicate primary keys and any relationship attributes.

- Each Department has a unique Department_ID, Name, and Office_Location.
- Each Department offers one or more Courses.
- Each Course has a unique Course_ID, Title, and Credits, and is offered by exactly one Department.
- Each Instructor has a unique Instructor_ID, Name, Email, and Phone, and works for exactly one Department.
- An Instructor can teach zero or more Courses, and each Course is taught by one or more Instructors.
- Each Student has a unique Student_ID, Name, Address, and Date_of_Birth.
- A Student can enroll in zero or more Courses, and each Course can have zero or more Students.
- For each Enrollment, store the Grade obtained by the student.

Student in Lab

Design a database for an emerging online bookstore. Based on interviews with the stakeholders, you have gathered the following requirements. Your task is to draw an **ER .**

1. Books and Authors

Each **Book** is identified by a unique **ISBN**. We also need to track its **Title**, **Price**, and the **Year** it was published.

An **Author** is recorded with a unique **Name**, an **Address**, and a **URL** (personal website).

Books are **written-by** authors. (Assume for this exercise that authors and books are connected in a way that allows multiple authors per book).

2. Publishers

Each **Publisher** has a unique **Name**, an **Address**, a **Phone** number, and a **URL**.

A book is **published-by** exactly one publisher.

3. Inventory and Warehousing

Books are stored in various **Warehouses**. Each warehouse has a unique **Code**, an **Address**, and a **Phone** number.

The relationship **stocks** connects books and warehouses. Crucially, we must keep track of the **Number** (quantity) of a specific book held in a specific warehouse.

4. Customers and Shopping Baskets

A **Customer** is identified by a unique **Email**. We also store their **Name**, **Address**, and **Phone** number.

Each customer can have a **Shopping-basket**. A basket is identified by a **basketID**.

The relationship **basket-of** connects a customer to their specific basket.

A basket **contains** books. We need to record the **Number** (quantity) of each book added to the basket.